Customer Purchase Intelligence: From Transactions to Strategy

1. Executive Summary

Retail businesses collect massive transaction data but often struggle to translate it into actionable decisions.
In this project, I analyzed **3,900 retail transactions** to uncover how customer demographics, purchasing habits, discounts, and subscriptions influence revenue and loyalty.

The project combines:

- Python for data preparation
- PostgreSQL for business analysis
- Power BI for stakeholder storytelling

The goal was to answer a simple question:

How can a retailer increase revenue and customer loyalty using data?

2. Business Problem

Retail managers needed clarity on:

- Who generates the most revenue?
- Do discounts actually increase spending?
- Which customers are likely to subscribe?
- Which products and categories should be promoted?
- Which customer segments should marketing focus on?

This project converts raw transactions into decision-ready insights.

3. Dataset Overview

The dataset contains **3,900 purchase records** with **18 variables** describing customers and transactions.

Key Data Areas

Customer profile

- Age
- Gender
- Location
- Subscription status

Purchase details

- Product and category
- Purchase amount
- Season, size, and color

Shopping behavior

- Discounts and promo usage
- Purchase frequency
- Shipping preference
- Product ratings

During exploration, **37 missing values** were found in the review rating column.

4. Data Preparation (Python)

1. Data Quality Assessment

Using pandas, I explored structure, data types, and summary statistics to identify inconsistencies and missing values.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discou Appli
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	39
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	22
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	N
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	N
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	N
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	N
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	N
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	N
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	N

Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
3900	3900	3900.000000	3900	3900
2	2	NaN	6	7
No	No	NaN	PayPal	Every 3 Months
2223	2223	NaN	677	584
NaN	NaN	25.351538	NaN	NaN
NaN	NaN	14.447125	NaN	NaN
NaN	NaN	1.000000	NaN	NaN
NaN	NaN	13.000000	NaN	NaN
NaN	NaN	25.000000	NaN	NaN
NaN	NaN	38.000000	NaN	NaN
NaN	NaN	50.000000	NaN	NaN

2. Handling Missing Ratings

Customer ratings are critical for product analysis.
Instead of removing rows, missing values were replaced with the **median rating within each product category** to preserve category-level trends.

This prevents bias while maintaining the dataset size.

3. Feature Engineering

To improve business analysis, new features were created:

Age groups

- Teen
- Young Adult
- Adult
- Senior

This enables demographic segmentation.

Purchase frequency metric
Converted purchase frequency into a consistent numerical format for analysis.

4. Removing Redundant Features

Promo code usage overlapped strongly with discount usage.
To avoid duplicated signals, the promo column was removed.

5. Database Integration

The cleaned dataset was loaded into PostgreSQL to enable structured business queries and scalable analysis.

5. Business Analysis in SQL

Instead of generic queries, the analysis was framed as **business questions**.

1. Which gender drives more revenue?

This analysis identifies the most valuable customer demographic.

Why it matters: Helps marketing allocate budget effectively.

	gender text	revenue numeric
1	Female	75191
2	Male	157890

2. Do discounts attract high spenders?

We identified customers who:

- used discounts
- still spent above average

Why it matters: Determines whether discounts attract bargain hunters or valuable customers.

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
11	24	88

Total rows: 839 Query complete 00:00:00.133

3. Which products delight customers the most?

Top products were ranked using average review ratings.

Why it matters: High-rated products are strong candidates for promotion.

	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

4. Does shipping speed influence spending?

Average purchase value was compared between:

- Standard shipping
- Express shipping

Why it matters: Shipping incentives may drive higher basket sizes.

	shipping_type text	round numeric
1	Standard	58.46
2	Express	60.48

5. Are subscriptions worth it?

Subscribers vs. non-subscribers were compared on the following:

- Average spend
- Total revenue

Why it matters: Measures the ROI of subscription programs.

	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

6. Which products rely heavily on discounts?

The products with the highest % of discounted purchases were identified.

Why it matters: Highlights potential margin risks.

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.66
3	Coat	49.07
4	Sweater	48.17
5	Pants	47.37

7. Customer Segmentation

Customers were grouped into:

- New (first purchase)
- Returning (2–5 purchases)
- Loyal (>5 purchases)

Why it matters: Enables lifecycle marketing strategies.

	customer_segment text	Number of Customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

8. Best-Selling Products by Category

The top three products per category were identified.

Why it matters: Guides inventory and campaign planning.

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessories	Jewelry	171
2	2	Accessories	Sunglasses	161
3	3	Accessories	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

9. Do repeat buyers subscribe more?

Examined whether frequent buyers are more likely to subscribe.

Why it matters: Identifies the best audience for subscription offers.

	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

10. Revenue by Age Group

Measured revenue contribution by demographic segment.

Why it matters: Enables targeted advertising.

	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

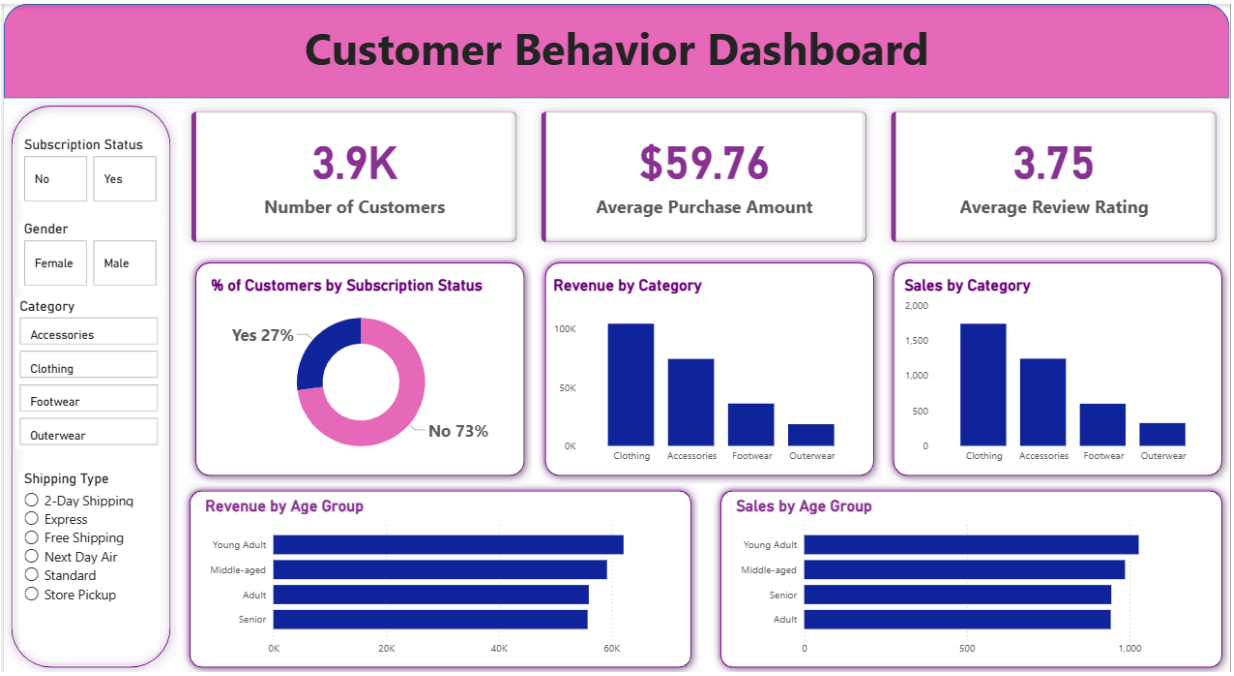
6. Dashboard Development (Power BI)

An interactive dashboard was created for non-technical stakeholders.

Key dashboard capabilities:

- Revenue breakdown by demographics
- Product and category performance
- Discount and subscription insights
- Customer segmentation visuals

The dashboard transforms SQL insights into decision-ready visuals.



7. Key Insights

Instead of generic bullet points, we'll phrase them like real insights:

- Subscribers generate significantly higher average order value.
- Express shipping customers spend more per purchase.
- Loyal customers show strong potential for subscription conversion.
- Several products rely heavily on discounts, indicating margin risk.
- Specific age groups contribute a disproportionate share of revenue.

8. Strategic Recommendations

1. Grow the Subscription Program

Target repeat buyers with subscription incentives and exclusive benefits.

2. Introduce a Loyalty Program

Reward returning customers to accelerate movement into the loyal segment.

3. Optimize Discount Strategy

Reduce blanket discounts and focus on products that convert without heavy promotions.

4. Promote High-Rated Products

Feature top-rated items in marketing campaigns to increase conversion rates.

5. Invest in High-Value Segments

Prioritize marketing toward demographics and shipping segments with higher spending behaviour.

9. Tools Used

- Python (Pandas)
- PostgreSQL
- Power BI

10. Project Outcome

This project illustrates the transformation of transaction data into a clear, actionable business strategy using a modern analytics stack.